Problem 1

A particle is confined in a one-dimensional infinite potential well of width L, with infinite potential outside the region $0 < x < L$. The normalized stationary states and energy levels are given by:

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \ldots$$

At time $t = 0$, the wavefunction of the particle is given by:

$$\psi(x, 0) = \mathcal{N} \cdot \left(\sin\left(\frac{\pi x}{L}\right) + \sin\left(\frac{3\pi x}{L}\right) + 2\sin\left(\frac{5\pi x}{L}\right)\right)$$

where $\mathcal{N}$ is a normalization constant.

Tasks:

1. Normalize the wavefunction and determine the value of $\mathcal{N}$.
2. Express $\psi(x, 0)$ in terms of the stationary states $\phi_n(x)$.
3. Write the time-dependent wavefunction $\psi(x, t)$.
4. Compute the probability of measuring each of the energies E_1, E_3, and E_5.
5. Determine whether $|\psi(x, t)|^2$ is time-dependent or not. Explain briefly.
6. Compute the expectation value of the energy $\langle \hat{H} \rangle$ in this state.

Solution

1. **Normalize the wavefunction:**

   Given:

   $$\psi(x, 0) = \mathcal{N}\left(\sin\left(\frac{\pi x}{L}\right) + \sin\left(\frac{3\pi x}{L}\right) + 2\sin\left(\frac{5\pi x}{L}\right)\right)$$

   We compute the normalization condition:

   $$\int_0^L |\psi(x, 0)|^2 \, dx = 1$$

   Using orthogonality of sine functions:

   $$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2}, \quad \int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = 0 \ (n \neq m)$$

So we have:

$$\int_0^L |\psi(x,0)|^2 dx = \mathcal{N}^2 \left(\frac{L}{2} + \frac{L}{2} + 4 \cdot \frac{L}{2} \right) = \mathcal{N}^2 \cdot 3L$$

Setting the integral to 1:

$$\mathcal{N}^2 \cdot 3L = 1 \quad \Rightarrow \quad \mathcal{N} = \sqrt{\frac{1}{3L}}$$

2. **Express $\psi(x,0)$ in terms of stationary states $\phi_n(x)$:**

   Using $\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$, we rewrite:

$$\psi(x,0) = \sqrt{\frac{1}{3L}} \left(\sin\left(\frac{\pi x}{L}\right) + \sin\left(\frac{3\pi x}{L}\right) + 2\sin\left(\frac{5\pi x}{L}\right) \right)$$

   Now we relate the sine terms to $\phi_n(x)$:

$$\psi(x,0) = \sqrt{\frac{1}{3}} \left(\frac{1}{\sqrt{2}}\phi_1(x) + \frac{1}{\sqrt{2}}\phi_3(x) + \frac{2}{\sqrt{2}}\phi_5(x) \right) = \sqrt{\frac{1}{6}}\phi_1(x) + \sqrt{\frac{1}{6}}\phi_3(x) + \sqrt{\frac{2}{3}}\phi_5(x)$$

3. **Time-dependent wavefunction $\psi(x,t)$:**

   Each stationary state evolves as $\phi_n(x)e^{-iE_n t/\hbar}$, so:

$$\psi(x,t) = \sqrt{\frac{1}{6}}\phi_1(x)e^{-iE_1 t/\hbar} + \sqrt{\frac{1}{6}}\phi_3(x)e^{-iE_3 t/\hbar} + \sqrt{\frac{2}{3}}\phi_5(x)e^{-iE_5 t/\hbar}$$

4. **Probabilities of measuring energies E_1, E_3, E_5:**

   The coefficients squared give the probabilities:

$$P(E_1) = \left| \sqrt{\frac{1}{6}} \right|^2 = \frac{1}{6}, \quad P(E_3) = \frac{1}{6}, \quad P(E_5) = \frac{2}{3}$$

5. **Is $|\psi(x,t)|^2$ time-dependent?**

   Yes, it is time-dependent. Since $\psi(x,t)$ is a superposition of multiple stationary states with different energies, the relative phases between them evolve in time. This leads to interference terms in $|\psi(x,t)|^2$ that depend on time.

6. **Expectation value of energy $\langle \hat{H} \rangle$:**

   The expectation value is:

$$\langle \hat{H} \rangle = \sum_n |c_n|^2 E_n = \frac{1}{6}E_1 + \frac{1}{6}E_3 + \frac{2}{3}E_5$$

   Using $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$, we compute:

$$\begin{aligned} \langle \hat{H} \rangle &= \frac{1}{6}E_1 + \frac{1}{6}E_3 + \frac{2}{3}E_5 = \frac{1}{6} \cdot E_1 + \frac{1}{6} \cdot 9E_1 + \frac{2}{3} \cdot 25E_1 \\ &= E_1 \left(\frac{1}{6} + \frac{9}{6} + \frac{50}{6} \right) = E_1 \cdot \frac{60}{6} = 10E_1 \end{aligned}$$

   So the expectation value is:

$$\langle \hat{H} \rangle = 10 \cdot \frac{\pi^2\hbar^2}{2mL^2}$$

Problem 2

A quantum system has a time-independent Hamiltonian with the following eigenstates and eigenvalues:

$$\hat{H}\,|+\rangle = E_+\,|+\rangle\,, \quad \hat{H}\,|-\rangle = E_-\,|-\rangle$$

with eigenvalues:

$$E_+ = \hbar\omega, \quad E_- = 2\hbar\omega$$

The initial state at time $t = 0$ is given by:

$$|\psi(0)\rangle = (1-i)\,|+\rangle + (2+i)\,|-\rangle$$

Tasks:

1. Normalize the initial state $|\psi(0)\rangle$.
2. Write down the time-evolved state $|\psi(t)\rangle$.

Solution

1. **Normalize the initial state $|\psi(0)\rangle$:**

   The unnormalized state is:

   $$|\psi(0)\rangle = (1-i)\,|+\rangle + (2+i)\,|-\rangle$$

   The normalization factor is computed from:

   $$\langle\psi(0)|\psi(0)\rangle = |1-i|^2 + |2+i|^2$$

   First compute each modulus squared:

   $$|1-i|^2 = 1^2 + 1^2 = 2, \quad |2+i|^2 = 2^2 + 1^2 = 4 + 1 = 5$$

   So:

   $$\langle\psi(0)|\psi(0)\rangle = 2 + 5 = 7$$

   The normalized state is:

   $$|\psi(0)\rangle = \frac{1-i}{\sqrt{7}}\,|+\rangle + \frac{2+i}{\sqrt{7}}\,|-\rangle$$

2. **Time-evolved state $|\psi(t)\rangle$:**

   Under unitary time evolution with a time-independent Hamiltonian:

   $$|\psi(t)\rangle = \frac{1-i}{\sqrt{7}} e^{-iE_+t/\hbar}\,|+\rangle + \frac{2+i}{\sqrt{7}} e^{-iE_-t/\hbar}\,|-\rangle$$

   Using $E_+ = \hbar\omega$, $E_- = 2\hbar\omega$, we obtain:

   $$|\psi(t)\rangle = \frac{1-i}{\sqrt{7}} e^{-i\omega t}\,|+\rangle + \frac{2+i}{\sqrt{7}} e^{-i2\omega t}\,|-\rangle$$

Problem 3

Consider a two-qubit system in the following entangled state:

$$|\psi\rangle_{AB} = \frac{1}{\sqrt{10}}\left(2\,|00\rangle + i\,|01\rangle + |10\rangle\right)$$

Tasks:

1. Normalize the state $|\psi\rangle_{AB}$ if necessary.
2. Write down the density matrix $\rho_{AB} = |\psi\rangle_{AB}\langle\psi|_{AB}$ in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.
3. Compute the reduced density matrix $\rho_A = \mathrm{Tr}_B(\rho_{AB})$, i.e., trace out Bob's qubit.
4. Determine whether ρ_A is a pure or mixed state. Justify your answer.

Solution

1. **Normalize the state $|\psi\rangle_{AB}$:**

   The given state is:

   $$|\psi\rangle_{AB} = \frac{1}{\sqrt{10}}\left(2\,|00\rangle + i\,|01\rangle + |10\rangle\right)$$

   Compute the norm:

   $$\||\psi\rangle_{AB}\|^2 = \frac{1}{10}\left(|2|^2 + |i|^2 + |1|^2\right) = \frac{1}{10}(4+1+1) = \frac{6}{10}$$

   So the state is *not* normalized. Normalize it:

   $$|\psi\rangle_{AB} = \frac{1}{\sqrt{6}}\left(2\,|00\rangle + i\,|01\rangle + |10\rangle\right)$$

2. **Write the density matrix $\rho_{AB} = |\psi\rangle_{AB}\langle\psi|_{AB}$:**

   Let us write $|\psi\rangle_{AB}$ in vector form using the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

   $$|\psi\rangle_{AB} = \frac{1}{\sqrt{6}}\begin{bmatrix}2\\ i\\ 1\\ 0\end{bmatrix}, \quad \rho_{AB} = |\psi\rangle\langle\psi| = \frac{1}{6}\begin{bmatrix}4 & 2i & 2 & 0\\ -2i & 1 & i & 0\\ 2 & -i & 1 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$$

3. **Compute the reduced density matrix $\rho_A = \mathrm{Tr}_B(\rho_{AB})$:**

   Recall that the reduced density matrix ρ_A is obtained by tracing over the second qubit (Bob's subsystem). We write the full density matrix in block form:

$$\rho_{AB} = \begin{bmatrix} \rho_{00,00} & \rho_{00,01} & \rho_{00,10} & \rho_{00,11} \\ \rho_{01,00} & \rho_{01,01} & \rho_{01,10} & \rho_{01,11} \\ \rho_{10,00} & \rho_{10,01} & \rho_{10,10} & \rho_{10,11} \\ \rho_{11,00} & \rho_{11,01} & \rho_{11,10} & \rho_{11,11} \end{bmatrix}$$

   Then:

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \begin{bmatrix} \rho_{00,00} + \rho_{01,01} & \rho_{00,10} + \rho_{01,11} \\ \rho_{10,00} + \rho_{11,01} & \rho_{10,10} + \rho_{11,11} \end{bmatrix}$$

   Using the matrix from part (2):

$$\rho_A = \frac{1}{6} \begin{bmatrix} 4+1 & 2+0 \\ 2+0 & 1+0 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$$

4. **Determine whether ρ_A is pure or mixed:**

   To determine if ρ_A is pure, compute $\mathrm{Tr}(\rho_A^2)$:

$$\rho_A^2 = \left(\frac{1}{6} \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} \right)^2 = \frac{1}{36} \begin{bmatrix} 25+4 & 10+2 \\ 10+2 & 4+1 \end{bmatrix} = \frac{1}{36} \begin{bmatrix} 29 & 12 \\ 12 & 5 \end{bmatrix}$$

$$\mathrm{Tr}(\rho_A^2) = \frac{1}{36}(29+5) = \frac{34}{36} = \frac{17}{18} < 1$$

   Therefore, ρ_A is a **mixed state**.

Problem 4

Consider the following normalized three-qubit state:

$$|\psi\rangle_{ABC} = \frac{1}{\sqrt{3}} \left(|001\rangle + |010\rangle + |100\rangle \right)$$

Tasks:

1. Compute the reduced density matrix $\rho_{AC} = \mathrm{Tr}_B(\rho_{ABC})$, i.e., trace out the middle qubit B.
2. Determine whether the reduced state ρ_{AC} is mixed or not.

Solution

1. **Compute the reduced density matrix $\rho_{AC} = \mathrm{Tr}_B(\rho_{ABC})$:**

   The given state is:

$$|\psi\rangle_{ABC} = \frac{1}{\sqrt{3}}\left(|001\rangle + |010\rangle + |100\rangle\right)$$

   The full density matrix is:

$$\begin{aligned}\rho_{ABC} = \frac{1}{3}\Big(&|001\rangle\langle 001| + |001\rangle\langle 010| + |001\rangle\langle 100| + \\ &|010\rangle\langle 001| + |010\rangle\langle 010| + |010\rangle\langle 100| + \\ &|100\rangle\langle 001| + |100\rangle\langle 010| + |100\rangle\langle 100|\Big)\end{aligned}$$

   To obtain ρ_{AC}, we trace over the middle qubit B. The partial trace over B means summing over $\langle b| \cdots |b\rangle$ where $b = 0, 1$. We examine each term:

   - Terms with $B = 0$:

$$\begin{aligned}\mathrm{Tr}_B(|100\rangle\langle 100|) &= |1\rangle_A |0\rangle_C \langle 1|_A \langle 0|_C = |10\rangle\langle 10| \\ \mathrm{Tr}_B(|010\rangle\langle 010|) &= |0\rangle_A |0\rangle_C \langle 0|_A \langle 0|_C = |00\rangle\langle 00| \\ \mathrm{Tr}_B(|010\rangle\langle 100|) &= |0\rangle_A |0\rangle_C \langle 1|_A \langle 0|_C = |00\rangle\langle 10| \\ \mathrm{Tr}_B(|100\rangle\langle 010|) &= |1\rangle_A |0\rangle_C \langle 0|_A \langle 0|_C = |10\rangle\langle 00|\end{aligned}$$

   - Terms with $B = 1$:

$$\mathrm{Tr}_B(|001\rangle\langle 001|) = |0\rangle_A |1\rangle_C \langle 0|_A \langle 1|_C = |01\rangle\langle 01|$$

   (Other cross terms like $|001\rangle\langle 010|$ or $|001\rangle\langle 100|$ trace to zero)

   Now summing all these contributions:

$$\rho_{AC} = \frac{1}{3}\left(|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + |00\rangle\langle 10| + |10\rangle\langle 00|\right)$$

   Write in matrix form using basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$\rho_{AC} = \frac{1}{3}\begin{bmatrix}1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0\end{bmatrix}$$

2. **Determine whether ρ_{AC} is mixed or not:**

   We compute $\mathrm{Tr}(\rho_{AC}^2)$ to check purity:

$$\rho_{AC}^2 = \left(\frac{1}{3}\begin{bmatrix}1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0\end{bmatrix}\right)^2 = \frac{1}{9}\begin{bmatrix}2 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0\end{bmatrix}$$

$$\mathrm{Tr}(\rho_{AC}^2) = \frac{1}{9}(2 + 1 + 2) = \frac{5}{9} < 1$$

   Therefore, ρ_{AC} is a **mixed state**.

Problem 5

Consider the following two-qubit quantum state:

$$|\psi\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle - |11\rangle\right)$$

The **Pauli-Y gate** is defined as:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Tasks:
Apply the operators $\mathbb{I} \otimes Y$ and $Y \otimes \mathbb{I}$ separately to the state $|\psi\rangle$, and compute the resulting two-qubit quantum states in the computational basis.

Solution

We are given the two-qubit state:

$$|\psi\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle - |11\rangle\right)$$

and the Pauli-Y gate:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

We will compute the result of applying $\mathbb{I} \otimes Y$ and $Y \otimes \mathbb{I}$ to $|\psi\rangle$ separately.

1. **Apply $\mathbb{I} \otimes Y$ to $|\psi\rangle$:**
   We apply the Y operator to the second qubit (right qubit) of each term:

$$\begin{aligned}
(\mathbb{I} \otimes Y)\,|00\rangle &= |0\rangle \otimes Y\,|0\rangle = |0\rangle \otimes i\,|1\rangle = i\,|01\rangle \\
(\mathbb{I} \otimes Y)\,|01\rangle &= |0\rangle \otimes Y\,|1\rangle = |0\rangle \otimes -i\,|0\rangle = -i\,|00\rangle \\
(\mathbb{I} \otimes Y)\,|10\rangle &= |1\rangle \otimes Y\,|0\rangle = |1\rangle \otimes i\,|1\rangle = i\,|11\rangle \\
(\mathbb{I} \otimes Y)\,|11\rangle &= |1\rangle \otimes Y\,|1\rangle = |1\rangle \otimes -i\,|0\rangle = -i\,|10\rangle
\end{aligned}$$

   Now act on the full state:

$$(\mathbb{I} \otimes Y)\,|\psi\rangle = \frac{1}{2}\left(i\,|01\rangle - i\,|00\rangle + i\,|11\rangle + i\,|10\rangle\right)$$

   Factor out i:

$$(\mathbb{I} \otimes Y)\,|\psi\rangle = \frac{i}{2}\left(-\,|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

2. **Apply $Y \otimes \mathbb{I}$ to $|\psi\rangle$:**
   We apply the Y operator to the first qubit (left qubit) of each term:

$$
\begin{aligned}
(Y \otimes \mathbb{I})\,|00\rangle &= Y\,|0\rangle \otimes |0\rangle = i\,|1\rangle \otimes |0\rangle = i\,|10\rangle \\
(Y \otimes \mathbb{I})\,|01\rangle &= Y\,|0\rangle \otimes |1\rangle = i\,|1\rangle \otimes |1\rangle = i\,|11\rangle \\
(Y \otimes \mathbb{I})\,|10\rangle &= Y\,|1\rangle \otimes |0\rangle = -i\,|0\rangle \otimes |0\rangle = -i\,|00\rangle \\
(Y \otimes \mathbb{I})\,|11\rangle &= Y\,|1\rangle \otimes |1\rangle = -i\,|0\rangle \otimes |1\rangle = -i\,|01\rangle
\end{aligned}
$$

Now act on the full state:

$$
(Y \otimes \mathbb{I})\,|\psi\rangle = \frac{1}{2}\left(i\,|10\rangle + i\,|11\rangle - i\,|00\rangle + i\,|01\rangle\right)
$$

Factor out i:

$$
(Y \otimes \mathbb{I})\,|\psi\rangle = \frac{i}{2}\left(-\,|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)
$$

Problem 6

Consider a five-qubit quantum system initialized in the state:

$$
|\psi\rangle = |+\rangle \otimes |0\rangle \otimes |1\rangle \otimes |0\rangle \otimes |1\rangle, \quad \text{where } |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)
$$

Tasks:
(1) A magnetic field acts only on the first qubit. The operation is described by the following unitary matrix:

$$
U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}
$$

Compute the resulting state $|\psi_1\rangle$ of the five-qubit system after applying $U \otimes \mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I}$ to the initial state $|\psi\rangle$.

(2) Now apply a controlled-NOT (CNOT) gate between the first and second qubits (first as control, second as target). Compute the new state $|\psi_2\rangle$ after this operation.

(3) Question: Is the final state **entangled**? If yes, identify which qubits are entangled.

Solution

We are given the initial five-qubit state:

$$
|\psi\rangle = |+\rangle \otimes |0\rangle \otimes |1\rangle \otimes |0\rangle \otimes |1\rangle, \quad \text{with } |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)
$$

Thus,

$$
|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \otimes |0\rangle \otimes |1\rangle \otimes |0\rangle \otimes |1\rangle = \frac{1}{\sqrt{2}}\left(|00101\rangle + |10101\rangle\right)
$$

(1) Apply unitary U to the first qubit only:

Given:

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \quad U|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle), \quad U|1\rangle = \frac{1}{\sqrt{2}}(i|0\rangle + |1\rangle)$$

Apply $U \otimes \mathbb{I}^{\otimes 4}$ to each term of the state:

$$U|0\rangle \otimes |0101\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + i|1\rangle\right) \otimes |0101\rangle = \frac{1}{\sqrt{2}}\left(|00101\rangle + i|10101\rangle\right)$$

$$U|1\rangle \otimes |0101\rangle = \frac{1}{\sqrt{2}}\left(i|0\rangle + |1\rangle\right) \otimes |0101\rangle = \frac{1}{\sqrt{2}}\left(i|00101\rangle + |10101\rangle\right)$$

So,

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00101\rangle + i|10101\rangle) + \frac{1}{\sqrt{2}}(i|00101\rangle + |10101\rangle)\right)$$

$$= \frac{1}{2}\left(|00101\rangle + i|10101\rangle + i|00101\rangle + |10101\rangle\right) = \frac{1}{2}\left((1+i)|00101\rangle + (1+i)|10101\rangle\right)$$

$$|\psi_1\rangle = \frac{1+i}{2}\left(|00101\rangle + |10101\rangle\right)$$

(2) Apply a CNOT with qubit 1 as control and qubit 2 as target:

Recall:

$$\mathrm{CNOT}_{1\to 2}|a\rangle|b\rangle = |a\rangle|b \oplus a\rangle$$

Apply CNOT to each term of $|\psi_1\rangle$:

$$\mathrm{CNOT}|00101\rangle = |01101\rangle, \quad \mathrm{CNOT}|10101\rangle = |11101\rangle$$

Thus:

$$|\psi_2\rangle = \frac{1+i}{2}\left(|01101\rangle + |11101\rangle\right)$$

(3) Is the final state entangled? If yes, which qubits are entangled?

We write:

$$|\psi_2\rangle = \frac{1+i}{2}\left(|1\rangle \otimes |1101\rangle + |0\rangle \otimes |1101\rangle\right) = \frac{1+i}{2}\left(|0\rangle + |1\rangle\right) \otimes |1101\rangle$$

This is a product state between the first qubit and the remaining four:

$$|\psi_2\rangle = \frac{1+i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \otimes |1101\rangle$$

Therefore, the final state is **not entangled**. It is a **separable** product state of the form:

$$|\psi_2\rangle = (\text{one-qubit state}) \otimes (\text{four-qubit state})$$

Problem 7

An electron is initially prepared in an eigenstate of the spin S_z, with eigenvalue $\frac{\hbar}{2}$. Please determine the possible outcomes when measuring the spin component S_x, and calculate the corresponding probabilities.

Solution

The electron is initially in an eigenstate of the spin operator S_z with eigenvalue $\frac{\hbar}{2}$, which corresponds to the spin-up state along the z-axis:

$$|+z\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

The spin component S_x has eigenstates:

$$|+x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

corresponding to eigenvalues $+\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$, respectively.

Step 1: Compute the probability amplitudes

We express $|+z\rangle$ in the S_x eigenbasis:

$$\langle +x| + z\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}$$

$$\langle -x| + z\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}$$

Step 2: Compute probabilities

The probabilities of measuring each eigenvalue of S_x are:

$$P\left(+\frac{\hbar}{2}\right) = |\langle +x| + z\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

$$P\left(-\frac{\hbar}{2}\right) = |\langle -x| + z\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

Conclusion:

The possible outcomes when measuring S_x are:

$$+\frac{\hbar}{2} \quad \text{and} \quad -\frac{\hbar}{2}$$

Each occurs with probability:

$$P = \frac{1}{2}$$

Problem 8

This spinor of an electron in the y-z plane of the Bloch sphere can be written as:

$$|\psi\rangle = \cos\left(\frac{\alpha}{2}\right)|0\rangle + i\sin\left(\frac{\alpha}{2}\right)|1\rangle, \quad \text{with } \alpha \in [0, \pi]$$

Please try to determine the possible outcomes when measuring the spin component S_y. Then, calculate the corresponding probabilities as functions of the parameter α.

Solution

We are given the spinor:

$$|\psi\rangle = \cos\left(\frac{\alpha}{2}\right)|0\rangle + i\sin\left(\frac{\alpha}{2}\right)|1\rangle, \quad \alpha \in [0,\pi]$$

This state lies in the y-z plane of the Bloch sphere. We are asked to compute the outcomes and probabilities of measuring the spin component S_y.

Step 1: Eigenstates of S_y

The spin operator $S_y = \frac{\hbar}{2}\sigma_y$, where:

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

The eigenstates of σ_y are:

$$|+y\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + i\,|1\rangle\right), \quad |-y\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - i\,|1\rangle\right)$$

corresponding to eigenvalues $+1$ and -1, hence $S_y = \pm\frac{\hbar}{2}$.

Step 2: Project $|\psi\rangle$ onto $|\pm y\rangle$

Compute the probability amplitudes:

$$\langle +y|\psi\rangle = \frac{1}{\sqrt{2}}\left(\langle 0| - i\,\langle 1|\right)\left(\cos\left(\frac{\alpha}{2}\right)|0\rangle + i\sin\left(\frac{\alpha}{2}\right)|1\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\alpha}{2}\right) - i^2\sin\left(\frac{\alpha}{2}\right)\right) = \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\alpha}{2}\right) + \sin\left(\frac{\alpha}{2}\right)\right)$$

$$\langle -y|\psi\rangle = \frac{1}{\sqrt{2}}\left(\langle 0| + i\,\langle 1|\right)\left(\cos\left(\frac{\alpha}{2}\right)|0\rangle + i\sin\left(\frac{\alpha}{2}\right)|1\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\alpha}{2}\right) + i^2\sin\left(\frac{\alpha}{2}\right)\right) = \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\alpha}{2}\right) - \sin\left(\frac{\alpha}{2}\right)\right)$$

Step 3: Compute probabilities

$$P\left(+\frac{\hbar}{2}\right) = |\langle +y|\psi\rangle|^2 = \frac{1}{2}\left(\cos\left(\frac{\alpha}{2}\right) + \sin\left(\frac{\alpha}{2}\right)\right)^2$$

$$= \frac{1}{2}\left(\cos^2\left(\frac{\alpha}{2}\right) + 2\cos\left(\frac{\alpha}{2}\right)\sin\left(\frac{\alpha}{2}\right) + \sin^2\left(\frac{\alpha}{2}\right)\right) = \frac{1}{2}\left(1 + \sin(\alpha)\right)$$

$$P\left(-\frac{\hbar}{2}\right) = |\langle -y|\psi\rangle|^2 = \frac{1}{2}\left(\cos\left(\frac{\alpha}{2}\right) - \sin\left(\frac{\alpha}{2}\right)\right)^2 = \frac{1}{2}\left(1 - \sin(\alpha)\right)$$

Conclusion:

The possible outcomes when measuring S_y are:

$$+\frac{\hbar}{2} \quad \text{with probability} \quad \frac{1}{2}(1+\sin\alpha)$$

$$-\frac{\hbar}{2} \quad \text{with probability} \quad \frac{1}{2}(1-\sin\alpha)$$

Problem 9

We consider a quantum entanglement swapping scenario involving four particles:

- Particle 1 is located in London,
- Particles 2 and 3 are in Shanghai,
- Particle 4 is in Sydney.

Initially, the entangled states are:

- Particles 1 and 2 share the following entangled state:

$$|\psi\rangle_{12} = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)$$

- Particles 3 and 4 are in a non-standard entangled state:

$$|\chi\rangle_{34} = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle - |11\rangle\right)$$

A Bell basis measurement is now performed on particles 2 and 3 in Shanghai.

Tasks:

1. Express the total initial state of the four particles $|\psi\rangle_{12} \otimes |\chi\rangle_{34}$ in terms of the Bell basis of particles 2 and 3.
2. After the Bell basis measurement on particles 2 and 3, determine the resulting state of particles 1 and 4 for each possible Bell outcome.
3. For each case, what local unitary operation should be performed on particle 1 (or particle 4) to convert the state of particles 1 and 4 into the maximally entangled Bell state $|\phi^+\rangle$?

Solution

Given:

- Particles 1 and 2 are in the Bell state:

$$|\psi\rangle_{12} = \frac{1}{\sqrt{2}}\left(|01\rangle_{12} - |10\rangle_{12}\right) = |\psi^-\rangle_{12}$$

- Particles 3 and 4 are in the state:

$$|\chi\rangle_{34} = \frac{1}{2}\left(|00\rangle_{34} + |01\rangle_{34} + |10\rangle_{34} - |11\rangle_{34}\right)$$

- A Bell basis measurement is performed on particles 2 and 3.

Step 1: Write total state $|\Psi\rangle_{1234} = |\psi^-\rangle_{12} \otimes |\chi\rangle_{34}$

We compute:

$$|\Psi\rangle_{1234} = \frac{1}{\sqrt{2}}(|01\rangle_{12} - |10\rangle_{12}) \otimes \frac{1}{2}(|00\rangle_{34} + |01\rangle_{34} + |10\rangle_{34} - |11\rangle_{34})$$

We write out all tensor products of particles 1–4:

$$|\Psi\rangle_{1234} = \frac{1}{2\sqrt{2}}\left[|0\rangle_1 |1\rangle_2 (|00\rangle + |01\rangle + |10\rangle - |11\rangle)_{34} - |1\rangle_1 |0\rangle_2 (|00\rangle + |01\rangle + |10\rangle - |11\rangle)_{34}\right]$$

$$= \frac{1}{2\sqrt{2}}\left[|0\rangle_1 (|1\rangle_2 |00\rangle_{34} + |1\rangle_2 |01\rangle_{34} + |1\rangle_2 |10\rangle_{34} - |1\rangle_2 |11\rangle_{34}) - |1\rangle_1 (|0\rangle_2 |00\rangle_{34} + |0\rangle_2 |01\rangle_{34} + |0\rangle_2 |10\rangle_{3}\right.$$

Group by qubits 2 and 3: We want to rewrite in the Bell basis of qubits 2 and 3:

$$|\phi^+\rangle_{23} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$
$$|\phi^-\rangle_{23} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle),$$
$$|\psi^+\rangle_{23} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle),$$
$$|\psi^-\rangle_{23} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Now express the full state $|\Psi\rangle_{1234}$ as a linear combination of Bell states of 2 and 3 and corresponding 1–4 states. This requires the standard Bell basis expansion trick:

$$|a\rangle_2 |b\rangle_3 = \sum_\beta c_\beta |\text{Bell}\rangle_{23} \otimes |\text{rest}\rangle_{14}$$

Instead of expanding everything explicitly, we can directly write the total state in the Bell basis of qubits 2 and 3 using the identity:

$$|\psi^-\rangle_{12} \otimes |\chi\rangle_{34} = \frac{1}{2} \sum_{\beta \in \{\phi^+, \phi^-, \psi^+, \psi^-\}} |\beta\rangle_{23} \otimes |\psi_\beta\rangle_{14}$$

where each $|\psi_\beta\rangle_{14}$ is obtained by partial swapping and rearrangement. After performing the computation, we find:

$$|\Psi\rangle_{1234} = \frac{1}{2}\Big(|\phi^+\rangle_{23} \otimes |\phi^-\rangle_{14} - |\phi^-\rangle_{23} \otimes |\phi^+\rangle_{14} + |\psi^+\rangle_{23} \otimes |\psi^-\rangle_{14} - |\psi^-\rangle_{23} \otimes |\psi^+\rangle_{14} \Big)$$

Answer to (1):

$$|\Psi\rangle_{1234} = \frac{1}{2}\left(|\phi^+\rangle_{23} \otimes |\phi^-\rangle_{14} - |\phi^-\rangle_{23} \otimes |\phi^+\rangle_{14} + |\psi^+\rangle_{23} \otimes |\psi^-\rangle_{14} - |\psi^-\rangle_{23} \otimes |\psi^+\rangle_{14}\right)$$

Answer to (2): After the Bell measurement on particles 2 and 3, the state of particles 1 and 4 collapses to:

- If result is $|\phi^+\rangle_{23}$, then $|\psi_{14}\rangle = |\phi^-\rangle_{14}$
- If result is $|\phi^-\rangle_{23}$, then $|\psi_{14}\rangle = -|\phi^+\rangle_{14}$
- If result is $|\psi^+\rangle_{23}$, then $|\psi_{14}\rangle = |\psi^-\rangle_{14}$
- If result is $|\psi^-\rangle_{23}$, then $|\psi_{14}\rangle = -|\psi^+\rangle_{14}$

Answer to (3): To convert each $|\psi_{14}\rangle$ into $|\phi^+\rangle_{14} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, the following local unitary operations are needed (on either particle 1 or 4):

- From $|\phi^-\rangle$: apply Z gate to particle 1 or 4.
- From $-|\phi^+\rangle$: apply global phase -1, or no physical action needed (overall phase).
- From $|\psi^-\rangle$: apply XZ to particle 1 (or 4).
- From $-|\psi^+\rangle$: apply X to particle 1 (or 4) and flip sign.

So, minimal local operations (on particle 1 or 4):

Bell Outcome on 2-3	Operation on Particle 1 to get $\|\phi^+\rangle_{14}$
$\|\phi^+\rangle$	Apply Z to particle 1
$\|\phi^-\rangle$	Do nothing (up to global phase)
$\|\psi^+\rangle$	Apply XZ to particle 1
$\|\psi^-\rangle$	Apply X to particle 1

Problem 10

We are given the following setup:

- Photons 2 and 3 (shared between Beijing and Shanghai) are in a partially entangled state with a hidden phase:
$$|\phi_{23}\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + e^{i\delta}|11\rangle\right)$$
- Photon 1 (in Beijing) is entangled with an ancilla qubit (photon 0), forming the state:
$$|\psi_{01}\rangle = \alpha|00\rangle + \beta|11\rangle$$
where photon 0 is kept in a quantum memory in Beijing.
- A joint Bell basis measurement is performed on photons 1 and 2.

Tasks:

1. Write down the total initial state $|\Psi\rangle_{0123} = |\psi_{01}\rangle \otimes |\phi_{23}\rangle$. Express it in terms of the Bell basis of photons 1 and 2.
2. For each possible outcome of the Bell measurement on photons 1 and 2, determine the resulting joint state of photons 0 and 3.

3. Suppose Bob (in Shanghai) receives classical communication about the Bell measurement outcome. What local operation should he perform on photon 3 to restore the entangled state $|\psi_{03}\rangle = \alpha|00\rangle + \beta|11\rangle$ between photon 0 and photon 3?

4. Discuss: if the phase δ is unknown, how does it affect the fidelity of the restored state? Can the entanglement still be used for reliable quantum teleportation or quantum key distribution?

Solution

Given:

- $|\psi_{01}\rangle = \alpha|00\rangle + \beta|11\rangle$
- $|\phi_{23}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + e^{i\delta}|11\rangle)$
- A Bell basis measurement is performed on qubits 1 and 2

(1) Total initial state $|\Psi\rangle_{0123}$

We write:

$$|\Psi\rangle_{0123} = (\alpha|00\rangle_{01} + \beta|11\rangle_{01}) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + e^{i\delta}|11\rangle_{23})$$

Tensor product:

$$|\Psi\rangle_{0123} = \frac{1}{\sqrt{2}}\left[\alpha|00\rangle_{01}|00\rangle_{23} + \alpha e^{i\delta}|00\rangle_{01}|11\rangle_{23} + \beta|11\rangle_{01}|00\rangle_{23} + \beta e^{i\delta}|11\rangle_{01}|11\rangle_{23}\right]$$

We rearrange qubit labels as (0)(1)(2)(3) and aim to express qubits 1 and 2 in the Bell basis:

$$|\phi^+\rangle_{12} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad |\phi^-\rangle_{12} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\psi^+\rangle_{12} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad |\psi^-\rangle_{12} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Now express $|\Psi\rangle_{0123}$ in the Bell basis of qubits 1 and 2:

We use the identity:

$$|ab\rangle = \frac{1}{2}\sum_{\mu}|\mu\rangle_{12} \otimes \sigma_\mu|ab\rangle$$

which implies:

$$|\Psi\rangle_{0123} = \frac{1}{2}\sum_{\mu\in\{\phi^+,\phi^-,\psi^+,\psi^-\}}|\mu\rangle_{12} \otimes |\psi^{(\mu)}\rangle_{03}$$

After performing the rearrangement and projecting onto the Bell basis of qubits 1 and 2, we find:

$$|\Psi\rangle_{0123} = \frac{1}{2}\Big(|\phi^+\rangle_{12} \otimes (\alpha |00\rangle_{03} + \beta e^{i\delta} |11\rangle_{03}) + |\phi^-\rangle_{12} \otimes (\alpha |00\rangle_{03} - \beta e^{i\delta} |11\rangle_{03})$$

$$+ |\psi^+\rangle_{12} \otimes (\alpha |01\rangle_{03} + \beta e^{i\delta} |10\rangle_{03}) + |\psi^-\rangle_{12} \otimes (\alpha |01\rangle_{03} - \beta e^{i\delta} |10\rangle_{03})\Big)$$

(2) Resulting state of photons 0 and 3 after Bell measurement:

- Outcome $|\phi^+\rangle_{12}$: $|\psi_{03}\rangle = \alpha |00\rangle + \beta e^{i\delta} |11\rangle$
- Outcome $|\phi^-\rangle_{12}$: $|\psi_{03}\rangle = \alpha |00\rangle - \beta e^{i\delta} |11\rangle$
- Outcome $|\psi^+\rangle_{12}$: $|\psi_{03}\rangle = \alpha |01\rangle + \beta e^{i\delta} |10\rangle$
- Outcome $|\psi^-\rangle_{12}$: $|\psi_{03}\rangle = \alpha |01\rangle - \beta e^{i\delta} |10\rangle$

(3) Operation on photon 3 to recover $\alpha |00\rangle + \beta |11\rangle$:
Bob needs to remove the effect of the Bell outcome and correct the phase $e^{i\delta}$.

- $|\phi^+\rangle$: apply $R_z(-\delta)$ on photon 3
- $|\phi^-\rangle$: apply $Z \cdot R_z(-\delta)$
- $|\psi^+\rangle$: apply $X \cdot R_z(-\delta)$
- $|\psi^-\rangle$: apply $XZ \cdot R_z(-\delta)$

Here, $R_z(-\delta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\delta} \end{pmatrix}$, which cancels the hidden phase.

(4) Discussion: Effect of unknown phase δ
If δ is unknown:

- The correction $R_z(-\delta)$ cannot be applied.
- As a result, the final state contains a phase mismatch between $|0\rangle$ and $|1\rangle$ components.
- The fidelity with respect to the desired entangled state $\alpha |00\rangle + \beta |11\rangle$ is reduced.
- This degrades the effectiveness of quantum teleportation and quantum key distribution.
- In particular, if δ fluctuates randomly, the entanglement becomes effectively useless due to decoherence.

Conclusion: The phase δ must be known or actively tracked/calibrated for reliable entanglement-based quantum communication. Otherwise, the fidelity of the shared entanglement is compromised.

Alternative Solution to question 2:

We project photons 1 and 2 onto $|\phi^+\rangle$:

$$\langle\phi^+|_{12}|\Psi\rangle_{0123} = \frac{1}{\sqrt{2}}\left(\langle 00| + \langle 11|\right)_{12}|\Psi\rangle_{0123}$$

Only the terms with $|00\rangle_{12}$ and $|11\rangle_{12}$ contribute:

$$= \frac{1}{\sqrt{2}}\left(\alpha|00\rangle_{03} + \beta e^{i\delta}|11\rangle_{03}\right)$$

Normalized:

$$|\psi_{03}^{(\phi^+)}\rangle = \alpha|00\rangle + \beta e^{i\delta}|11\rangle$$

Bell state: $|\phi^-\rangle_{12} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

$$\langle\phi^-|_{12}|\Psi\rangle_{0123} = \frac{1}{\sqrt{2}}\left(\alpha|00\rangle_{03} - \beta e^{i\delta}|11\rangle_{03}\right) \Rightarrow |\psi_{03}^{(\phi^-)}\rangle = \alpha|00\rangle - \beta e^{i\delta}|11\rangle$$

Bell state: $|\psi^+\rangle_{12} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

Only the terms with $|01\rangle_{12}$ and $|10\rangle_{12}$ contribute:

$$\langle\psi^+|_{12}|\Psi\rangle_{0123} = \frac{1}{\sqrt{2}}\left(\alpha|01\rangle_{03} + \beta e^{i\delta}|10\rangle_{03}\right) \Rightarrow |\psi_{03}^{(\psi^+)}\rangle = \alpha|01\rangle + \beta e^{i\delta}|10\rangle$$

Bell state: $|\psi^-\rangle_{12} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

$$\langle\psi^-|_{12}|\Psi\rangle_{0123} = \frac{1}{\sqrt{2}}\left(\alpha|01\rangle_{03} - \beta e^{i\delta}|10\rangle_{03}\right) \Rightarrow |\psi_{03}^{(\psi^-)}\rangle = \alpha|01\rangle - \beta e^{i\delta}|10\rangle$$

Summary: After the Bell measurement on photons 1 and 2, the state of photons 0 and 3 becomes:

- $|\phi^+\rangle_{12} \Rightarrow \alpha|00\rangle + \beta e^{i\delta}|11\rangle$
- $|\phi^-\rangle_{12} \Rightarrow \alpha|00\rangle - \beta e^{i\delta}|11\rangle$
- $|\psi^+\rangle_{12} \Rightarrow \alpha|01\rangle + \beta e^{i\delta}|10\rangle$
- $|\psi^-\rangle_{12} \Rightarrow \alpha|01\rangle - \beta e^{i\delta}|10\rangle$